

HydroSkin: Rapid Prototyping On-Skin Interfaces via Low-Cost Hydrographic Printing

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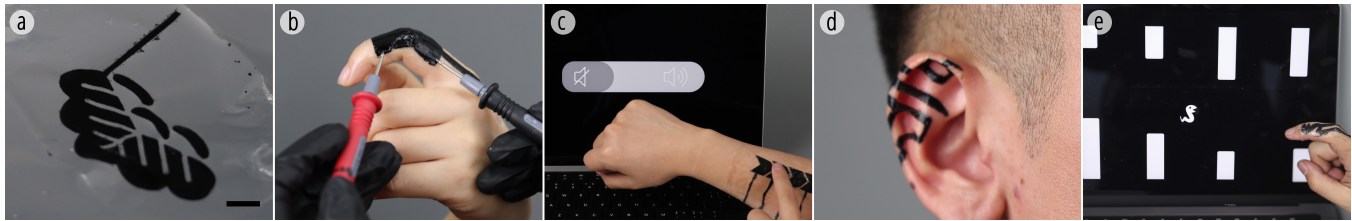


Figure 1: (a) Manufacturing process of HydroSkin. (b) Testing conductivity performance of low-cost mixed ink. (c) Arm slider controller for volume control. (d) Ear touch controller for quick phone call answering. (e) Finger game controller for controlling up and down movement of snake icon. (Scale bar: 5 mm.)

ABSTRACT

We introduce HydroSkin, a low-cost, time-efficient, and accessible fabrication technique for on-skin interfaces. By mixing carbon ink with sodium alginate, we developed a low-cost conductive material suitable for hydrographic printing technology. Compared to existing fabrication methods, our approach streamlines the fabrication process and utilizes cost-effective ink without requiring specialized skills. We showcased multiple application scenarios to demonstrate how HydroSkin can enable users to customize the

design and quickly fabricate their on-skin interfaces, including a finger game controller, an ear toucher, an arm control slider, and a fatigue monitoring mask.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*.

KEYWORDS

On-skin Interface, Fabrication, Electronic Skin, Low-cost, Hydrographic Printing

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1 INTRODUCTION

Personal fabrication for on-skin interfaces is rapidly emerging as a significant aspect of mobile computing, offering the Human-Computer Interaction (HCI) community a wide array of methods and materials. Researchers have been exploring various techniques for personal fabrication of flexible on-skin interfaces. For instance, they have utilized silica gel [17] with laser cutting to develop conformal on-skin interfaces. The choice of materials and workflow in this domain is heavily influenced by the desired functionality and display properties of the on-skin interfaces [4, 8, 9, 11, 18].

In the current field, the main challenges are the amount of time required, the cost of fabrication, and the accessibility of materials. However, the majority of personal fabrication methods have not yet met the demand for creating flexible surfaces with minimal effort and cost. Prior works prefer traditional approaches [4], such as using PVA film and screen printing techniques to create on-skin interfaces. For instance, producing SkinMarks takes 3-3.5 hours, with the screen printing mask taking about 2.5 hours, and fabricating all layers of a display tattoo takes nearly an hour [18].

We contribute a new fabrication technology for flexible on-skin interfaces that are created with carbon ink, which is available in our daily life. Specifically, we apply carbon ink and tape masks to optimize the original hydrographic printing, which are easily accessible in stationery stores, avoiding the long hours of screen printing. And according to the test results, we have reduced the time by almost 10 times in the process of fabrication in units of 4 cm × 4 cm. In this way, we hope to provide users with a higher degree of customization and more exploration possibilities for on-skin interfaces in the HCI community. HydroSkin enables users to easily create their own sensing components based on their demands (Figure 1). Given the skills and demand of users, we offer two options: free creation and high-precision creation. Whether it is directly using carbon ink for hand drawing or using laser cutting for cutting and then printing, on-skin interfaces with certain functionality can be obtained. In summary, the contributions of this paper include:

- A rapid fabrication method for creating thin, flexible, and body-conformal interfaces through hydrographic printing.
- A low-cost conductive composite ink material that enables skin-safe, functional, and highly customizable circuits.
- Technical evaluations of HydroSkin, including compatibility, robustness, efficiency, and cost.
- HydroSkin Applications that demonstrate the low threshold for users to create functional and aesthetic on-skin interfaces.

2 RELATED WORK

This work builds on research on on-skin interfaces, rapid fabrication for interaction surfaces and conductive materials.

2.1 On-skin Interfaces

The human body is a highly potential interactive interface, with on-skin interfaces and wearable devices being two approaches for exploration [10]. HCI community has demonstrated that on-skin interfaces hold immense potential for touch interaction beyond rigid touch screen. On-skin interface is a promising input-output interface, characterized by its highly flexible and comfortable nature, as well as its multifunctional capabilities [15]. HCI Researchers have

explored the technology and the fabricating processes of skin interfaces to enable users to create prototype designs. HCI researchers are primarily exploring the materials [3, 6], manufacturing processes [17, 19], and technologies [14] used in on-skin interfaces, and are committed to providing users with a richer interactive experience. For example, Some explore materials like memory alloys to make skin interfaces more adaptable to the human body [5]. Others are dedicated to exploring the complexity of human interface circuits and the greater possibilities of interaction [7]. While versatile, most previous works rely on either expensive equipment and materials [12, 16] or tedious handcraft. We aim to lower the barrier to customization and fabrication to support conformal on-skin interfaces.

2.2 Rapid Fabrication for Flexible Interfaces

Flexible on-skin interfaces typically involve specialized skills, high costs materials, and time-consuming processes. Prior researches have mainly focused on adhesion and conductivity properties in skin interfaces [13]. With limited attention to simplifying and cost-effectively streamlining the fabrication process, The manufacturing process and barriers to entry for on-skin interfaces present certain difficulties for the average user [19]. Traditional mask printing usually takes over 3 hours, mainly due to 2 hours required for silk-screen printing [2, 4, 18]. To address this issue, we replaced silk-screen printed masks with laser-cut masks, reducing the entire process, including the drying phase, to just 3 to 5 minutes, while maintaining similar edge quality for improved pattern clarity.

2.3 Conductive Materials

Recent advancements in low-cost on-skin fabrication have highlighted the use of ubiquitous materials [8, 9, 18]. Despite the trend towards bio-materials and composite materials for their enhanced functionalities, their availability remains a concern. For example, the silver ink used in [4] is unusual and hard to find. Our study leverages everyday materials to simplify the creation of on-skin interfaces and to lower barriers in sensor technology. Carbon ink emerges as a particularly accessible and cost-effective choice, costing only a few cents, and is notable for its biocompatibility and strong conductivity, making it suitable for sensor applications. The Do-It-Yourself approach also employs low-cost Carbon Black, yet incorporates a diverse range of mixed materials in the manufacturing process, such as Gelatin and Alginate [9]. Moreover, the preparation process of these raw materials is time-consuming. Figure 2 offers a comparative overview of the materials used and the fabrication methods.

3 HYDROSKIN METHOD

Leveraging hydrographic printing technology [1], HydroSkin employs low-cost carbon ink mixed with Sodium alginate thickener as a functional medium, utilizing water transfer printing to apply it onto human skin for sensory purposes. This approach extends the applicability of conventional electronic devices across various geometries and materials, achievable with only basic and readily available equipment and materials.

Example	Time	Cost	Accessibility of Materials
ObjectSkin[4]	> 3 h	> = \$ 50	lab material (silver ink, off sale)
SkinMarks[18]	3~3.5 h	> = \$ 50	lab material (PEDOT)
DuoSkin[8]	>10 min	< \$ 2.5	off the shelf (gold leaf)
Lotio[15]	>10 min	> = \$ 50	everyday material (lotion)
HydroSkin	< 10 min	< \$ 1	everyday material (carbon ink)

■ We estimate the cost based on the pattern size: 4 cm × 4 cm

Figure 2: Comprehensive comparison of on-skin interface interaction examples.

3.1 Materials

As shown in Figure 3, HydroSkin is composed of three layers of materials: water transfer film, functional ink, and protective coating.

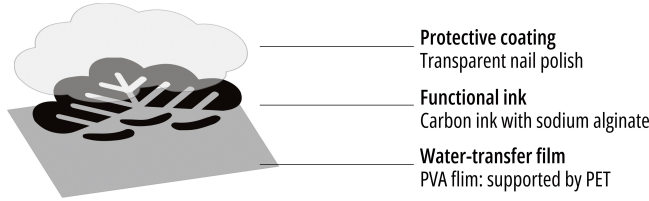


Figure 3: HydroSkin is composed of three layers of materials.

3.1.1 Transfer Substrate. We selected a water transfer film commonly used in industrial applications (YiYin brand, costing \$0.41 per A4 size) as the transfer substrate. This film consists of a composite of PET and PVA films, with PET facilitating protection and PVA handling transfer. The widespread use of PVA film in hydrographic printing is attributed to its excellent water solubility, eco-friendliness, uniform transfer capabilities, and versatility with various materials. Additionally, PVA film’s curvature adaptability and skin-friendliness make it a preferred choice for HydroSkin.

3.1.2 Functional Ink. We chose low-cost carbon ink (Hero brand, \$0.93 for 70 mL) mixed with 1% sodium alginate as the functional ink material, based on the following considerations: 1) The ink should be commonly used and familiar to ensure affordability and broad acceptability; 2) It needs to exhibit good conductivity; 3) Compatibility with water transfer film is essential.

To verify these characteristics, we conducted an electrical resistance test on carbon ink mixed with different concentrations of sodium alginate. We tested its conductivity on rectangular patterns that were 2 cm wide and increased in length from 1 to 6 cm (with a sample every 1 cm). As shown in Figure 4a-b, our experiments revealed that omitting or using insufficient sodium alginate leads to the ink cracking upon drying on the water transfer film, thereby significantly increasing electrical resistance. Conversely, an overly high concentration of sodium alginate causes uneven printing and delays the drying process. Carbon ink mixed with 1% concentration

of sodium alginate proved to be the optimal functional ink. In addition, to assess the impact of functional ink thickness on electrical performance, we evaluated the resistance with different printing cycles (Figure 4c). The experiment demonstrated that printing with the roller more than four times is sufficient to ensure good electrical performance.

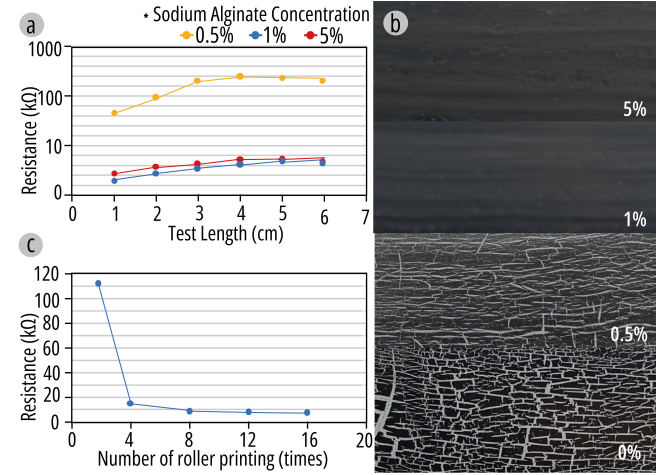


Figure 4: The (a) electrical resistance and (b) compatibility performance of carbon ink mixed with sodium alginate of different concentrations show significant differences on the water transfer film. (c) The electrical resistance of functional ink with different numbers of roller printing.

3.1.3 Protective Coating. Considering the water-soluble nature of carbon ink, we have judiciously selected transparent nail polish (MHBH brand, \$0.42 for 13 mL)—a widely accessible and well-known material—as a protective coating. This choice proficiently mitigates the blurring of edges and dissolution of patterns throughout the water transfer process.

3.2 Fabrication Pipeline

Figure 5 shows the fabrication pipeline of HydroSkin, demonstrating its advantages in rapid prototyping.

3.2.1 Painting Process. The painting process entails the application of personalized patterns onto water transfer films. We use carbon-based ink mixed with sodium alginate as functional ink to inscribe the required pattern onto the PVA side of the film. Specifically, pattern masks are precisely cut from adhesive tape using a laser cutting machine or paper cutter. Functional ink is then evenly applied to the template using a roller. Rapid drying of the ink is achieved using a hairdryer. Once the ink has dried, the adhesive tape is carefully removed, and a thin layer of transparent nail polish is applied as a protective coating to ensure durability.

The above method can create transfer patterns with more delicate edges and a more digitalized appearance. We also provide a method of hand-drawing patterns using brushes, which is significantly simpler and faster, providing enhanced flexibility for personalized and aesthetically driven customization.

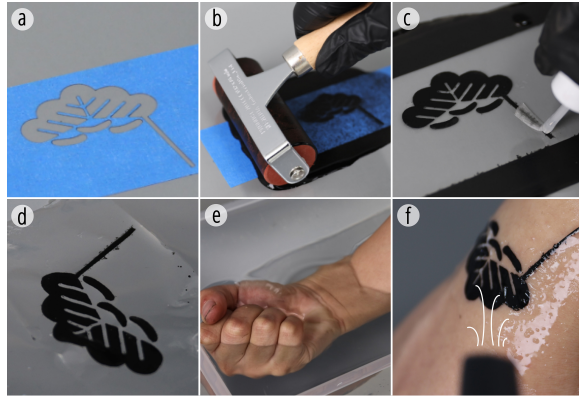


Figure 5: HydroSkin's Two-Step Process: (a-c) **Painting:** Creating skin circuit masks using graphic design software, followed by fabrication involving the application of low-cost ink mixed with sodium alginate for conductivity and a protective nail polish layer. (d-f) **Transfer:** Executing water transfer, completing the overall application, and drying with a hairdryer.

3.2.2 Transferring Process. In the subsequent step of the water transfer printing process, the PET film is first removed from the water transfer film. The PVA film is then laid flat in water, maintained at a temperature between 25 to 35°C, for approximately 15-30 seconds. Once the film softens, align the body parts with the pattern and immerse them in water. The immersion angle is meticulously chosen to correspond with the contour or texture of the processed surface, aiding in the efficient expulsion of air. During the transfer, gently stir the object underwater for 3 to 5 seconds to facilitate the removal of the non-patterned areas of the PVA film. After completing the transfer, a hair dryer can be effectively used for rapid drying.

Traditional industrial processes often use activators to enhance adhesion and stretchability of transfer films. However, HydroSkin, designed for human skin application, omits this step. Results have shown that omitting activators has no significant impact on transfer effectiveness.

4 APPLICATIONS

4.1 Finger Game Controller

We developed an innovative game controller based on finger movements, ingeniously using HydroSkin technology to transfer a snake pattern onto the finger (Figure 6). By translating the bending of the finger into changes in the pattern's resistance, it allows for controlling the movement of the snake in the game through finger gestures. By blending physical interaction with digital gameplay, this method significantly enriches the gaming experience. It lets players use simple finger movements to control their game characters, making the action both natural and intuitive.

4.2 Ear Toucher

We designed a touch sensor for the ear (Figure 7), allowing users to swiftly respond with a simple touch during situations where taking calls is inconvenient, like in meetings. The pattern connects

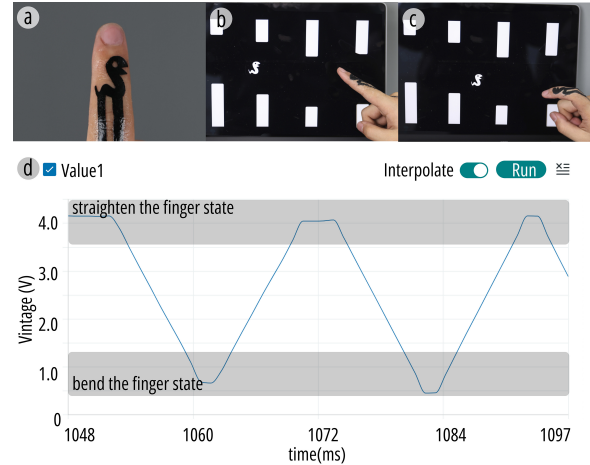


Figure 6: (a) Snake pattern on the finger featuring bend sensing and monitoring. (b-c) Finger bending controls the snake's upward and downward movements in the game. (d) Record the pressure sensor signal displayed on the Arduino IDE.

to a touch-sensitive module (MPR121) for touch capacitance sensing. The application uses hand-painted patterns to avoid the use of laser cutting machines or other equipment, further reducing the threshold of personalized fabrication. It demonstrates its adaptability to the complex concave and convex surfaces of the human body through hydrographic printing technology, showcasing its vast potential in on-skin interfaces.

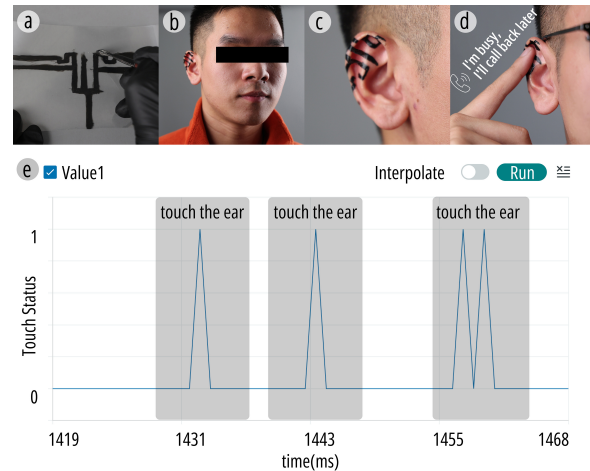


Figure 7: (a) Hand-drawn transfer pattern. (b-c) Transfer the pattern onto the ear. (d) After lightly touching the ear with your finger, quickly reply to the call. (e) Record the capacitance sensor signal displayed on the Arduino IDE.

4.3 Arm Control Slider

We developed a touch slider on the arm skin interface for adjusting the volume level (Figure 8). It operates on a principle similar to the ear toucher. Each arrow pattern serves as a capacitive touch button, each corresponding to different volume settings. The touch

slider is connected to a capacitive touch module (MPR121), utilized to detect the touch status. Compared to single-touch control, this single-column slider provides enhanced precision in adjustments.

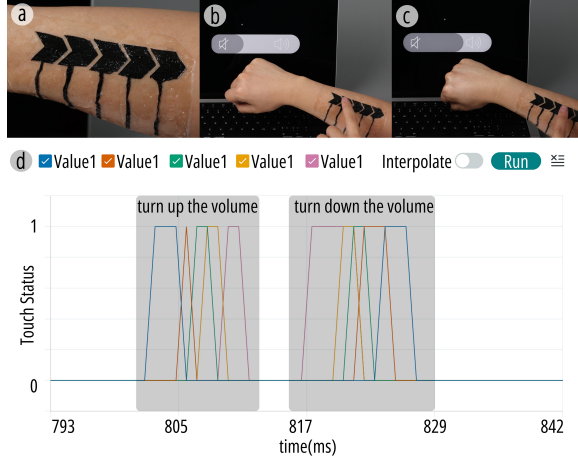


Figure 8: (a) Arm interface featuring a sliding controller, (b-c) utilizing capacitive touch sensing for volume adjustment. (d) Record the capacitance sensor array signals displayed on the Arduino IDE.

5 EVALUATION

5.1 Abrasion Resistance of HydroSkin

To evaluate the practicality of HydroSkin for daily use, we carried out bend tests. On the upper side of the index finger, we transferred a rectangular pattern. Figure 9 demonstrates that after 100 bends, the pattern maintained its basic integrity, showing no signs of fractures, cracks, or significant displacement, and the electrical resistance varied by less than 8%. Drawing from these testing insights, we suggest that during the manufacturing process, the skin should be as stretched as possible for transferring (such as submerging a curled finger into water) to better ensure the preservation of HydroSkin’s functionality.

5.2 Efficiency and Cost of HydroSkin

Given that HydroSkin is intended for daily wear and decoration on the human body, its transfer efficiency and cost are crucial design considerations. As illustrated in Figure 10, the consumption time showcases the production efficiency of the application case discussed in Section 4. These designs can be quickly transferred and formed. Cost estimation primarily takes into account the conductive ink and water transfer film. Since the cost of electronic component modules can be offset by their reusability, their one-time expenditure is negligible.

6 LIMITATIONS AND DISCUSSIONS

6.1 Alignment Precision

We have adopted manual alignment and simplified immersion techniques to support rapid transfer fabrication. Although the error

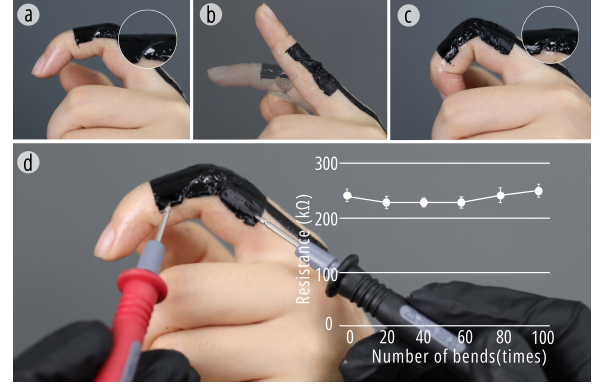


Figure 9: (a) Initial surface condition, (b) after 100 bends, (c) observing changes in surface texture, and (d) conducting resistance testing.

Applications	Time			Cost
	paint time	transfer time	total time	functional material
Finger game controller	3 min	1 min	4 min	< \$ 0.12
Ear toucher	2 min	1 min	3 min	< \$ 0.10
Arm control slider	2 min	1 min	3 min	< \$ 0.20

Figure 10: Production time and cost evaluation of HydroSkin application cases, the total time also encompasses the duration for painting and the drying process of transfer.

from manual alignment may accumulate as the pattern size increases, this type of error is still acceptable within the adaptability range of most prototypes. Moreover, the precise positioning of the pattern relies on manual operation, and its accuracy largely depends on the production experience of the operator. If a more accurate computational model [21] is adopted for guiding the production of HydroSkin, it will be possible to achieve precise positioning of the pattern.

6.2 Washability and Wearability

By harnessing the unique properties of PVA film, a water-transfer material, users can effortlessly refresh their skin decorations. A simple rinse and gentle rub allow for swift changes, accommodating diverse occasions and styles seamlessly. However, in some scenarios (such as exercise sweating or daily washing), the patterns tend to fall off, limiting its duration on the skin to no more than a day. Additionally, the nail polish coating may cause a sensation of tightness after HydroSkin has dried. In response, our future direction includes developing a comfortable, waterproof protective layer. We plan to conduct workshops to invite users to participate in and evaluate their actual HydroSkin experiences [18].

6.3 Procedure Limitation

HydroSkin’s current transfer process requires dipping body parts in water. This method does not align well with traditional body

decoration habits. It is also not suitable for transferring designs to large skin areas like the chest or back. These limitations restrict HydroSkin's use on large body areas. In the future, we plan to develop a new type of tattoo paper. This paper will be suitable for larger areas and complex shapes, without needing water immersion. This advancement will simplify the process, making it more natural and convenient. It will also expand the application possibilities.

6.4 High-quality Interconnection

The current application case connects functions to an Arduino board via fine copper wires, using Bluetooth for wireless communication, but remains lab-bound. Future exploration could involve synchronously transferring metal circuits and electronic components onto complex surfaces [20], or integrating transferred patterns with metallic accessories (e.g., earrings, bracelets) and miniaturized circuit boards (e.g., Seeeduino).

7 CONCLUSION

In this paper, we introduce HydroSkin, a low-cost, time-efficient, and accessible sensing fabrication technique using carbon ink for on-skin interfaces. The primary contribution of the paper is the provision of cost-effective and readily available materials, along with the proposal of a swift transfer fabrication process with no barriers. This democratizes and makes functional skin decorations more common and accessible to a wider audience. In the future, we aim to enhance the functionality of HydroSkin, enabling richer sensory and display capabilities on the skin. Furthermore, we will continually refine the fabrication process to accommodate various application scenarios, ensuring user-friendly transfer processes that cater to everyday decorative needs.

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